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ABSTRACT

WAVIS (Wildlife Acoustic Vocalization Identification System) is an AI-based system designed to identify animal species from environmental audio and estimate their approximate distance using a single microphone input. In dense forest environments, visual detection of animals is often difficult due to vegetation and limited visibility. This project leverages deep learning and audio signal processing to enable real-time wildlife understanding through sound.

The system converts raw audio signals into Mel spectrogram representations and uses deep convolutional neural networks for classification across 258 wildlife sound classes. Additionally, heuristic-based acoustic analysis is used to estimate relative distance and directional cues. A multi-agent architecture is implemented to handle sound acquisition, feature extraction, classification, and decision fusion in parallel.

Experimental results demonstrate that the model achieves a Top-1 accuracy of approximately 51% and Top-5 accuracy of over 72%, showing strong performance given the complexity of the dataset. The system provides real-time outputs including species identification and approximate spatial context (left/right, near/far, above/ground).

This work highlights the potential of AI in wildlife monitoring, eco-tourism, and environmental conservation by enabling intelligent acoustic sensing using low-cost hardware.

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CHAPTER 1

INTRODUCTION

Wildlife ecosystems are rich, complex environments where many animal species rely heavily on acoustic communication for survival, mating, territorial signaling, and predator warnings. In such environments, visual identification of animals is often challenging due to dense vegetation, low visibility, and the natural behavior of animals to remain hidden. As a result, sound becomes one of the most reliable indicators of animal presence and activity.

With the rapid advancement of **Artificial Intelligence (AI)** and **Machine Learning (ML)**, especially in the domain of audio signal processing, it has become possible to automatically analyze environmental sounds and extract meaningful information from them. Deep learning models, particularly Convolutional Neural Networks (CNNs), have shown remarkable performance in classifying complex audio patterns when combined with appropriate feature representations such as Mel spectrograms.

This project, **WAVIS (Wildlife Acoustic Vocalization Identification System)**, aims to bridge the gap between acoustic data and human understanding by providing an intelligent system capable of identifying animal species from environmental audio in real time. The system processes raw audio signals, converts them into frequency-based representations, and uses a trained deep learning model to classify the sound into one of the many wildlife species.

The uniqueness of this project lies in its ability to function using a single microphone, making it highly portable and accessible. Unlike traditional localization systems that require multiple sensors, WAVIS provides approximate spatial awareness through intelligent feature interpretation..

Overall, WAVIS represents a step forward in leveraging AI for environmental intelligence, offering a smart, real-time, and user-friendly solution for wildlife acoustic identification and interaction.

1.1 Background

Wildlife monitoring has traditionally been conducted using visual observation methods such as direct surveys, camera traps, and manual tracking. While effective, these methods are limited in dense environments where animals are difficult to spot. Acoustic monitoring has emerged as an alternative approach, as many animals produce distinct vocalizations that can be used for identification.

Early acoustic analysis relied on manual listening and basic signal processing techniques. With the development of digital signal processing, methods such as **Fast Fourier Transform (FFT)** and **Short-Time Fourier Transform (STFT)** were used to extract frequency information from audio signals. Features like **Mel-Frequency Cepstral Coefficients (MFCCs)** further improved classification performance.

The introduction of deep learning significantly enhanced the capability of audio classification systems. Convolutional Neural Networks (CNNs) can process spectrogram images and automatically learn important features, resulting in improved accuracy compared to traditional methods.

However, most existing systems focus only on classification and often require advanced hardware or multiple microphones for localization. Additionally, many systems are designed for offline processing rather than real-time applications.

The WAVIS system builds on these advancements by combining deep learning with a multi-agent architecture to enable real-time species identification and approximate distance estimation using a single microphone. This makes it more practical and accessible for real-world usage.

1.2 Motivation

The motivation for developing WAVIS arises from the limitations of current wildlife observation methods and the need for smarter, more accessible solutions. In forest environments, animals are often heard before they are seen. However, most people cannot identify species based on sound alone, reducing the effectiveness of wildlife exploration and learning.

For tourists and nature enthusiasts, recognizing animal sounds can significantly enhance the experience by providing better awareness and interaction with the environment. A real-time system that identifies species from sound can act as an intelligent guide.

From a research perspective, wildlife monitoring is essential for biodiversity conservation. Traditional methods are labor-intensive and sometimes intrusive. Acoustic monitoring offers a non-invasive alternative, but existing systems are often complex, expensive, or not suitable for real-time use.

Advancements in deep learning and audio processing provide an opportunity to build efficient and accurate systems. Tools such as PyTorch and Librosa make it possible to develop models capable of handling complex environmental audio.

Another key motivation is to design a system that works with minimal hardware. Most localization techniques require multiple microphones, which increases cost and complexity. WAVIS aims to provide approximate spatial information using a single microphone, making it portable and practical.

Thus, the project is motivated by the need to create a real-time, AI-powered system that enhances wildlife understanding while remaining accessible and efficient.

1.3 Objectives

The primary objective of the WAVIS project is to develop an intelligent system that can identify wildlife species from environmental audio signals and provide additional contextual information such as distance and direction.

1. The first objective is to build a deep learning-based classification model capable of identifying multiple animal species. This involves collecting and preprocessing audio data and converting it into Mel spectrograms for effective feature representation.
2. The second objective is to implement efficient audio feature extraction techniques. Methods such as Short-Time Fourier Transform (STFT) and Mel scaling are used to capture both temporal and frequency characteristics of the audio signal.
3. Another objective is to estimate the relative distance of the sound source. This is achieved by analyzing acoustic features such as amplitude and signal energy to categorize the source as near, medium, or far.
4. The project also aims to design a multi-agent system architecture where different components such as audio acquisition, preprocessing, classification, and decision-making operate independently but collaboratively. This improves modularity and system efficiency.
5. Additionally, the system should support real-time processing, allowing users to receive immediate predictions. This requires optimizing both the model and the data pipeline.
6. Finally, the project aims to develop a simple and user-friendly interface that displays the predicted species and contextual information, making the system useful for tourists, researchers, and conservationists.

CHAPTER 2

LITERARY REVIEW

Wildlife acoustic monitoring has become an important area of research for studying biodiversity and animal behavior in a non-invasive manner. Early approaches relied on manual audio recording and expert-based identification, where specialists analyzed sound patterns to identify species. Although effective, this method was time-consuming and not scalable for large datasets (Chen et al., 2023) [9].

To automate this process, traditional signal processing techniques were introduced. Methods such as Short-Time Fourier Transform (STFT) were used to convert audio signals into frequency representations, enabling feature extraction from time-varying signals (Griffin & Lim, 1984) [7]. Additionally, perceptual scaling techniques like the Mel scale were developed to better represent human auditory perception of sound frequencies (Stevens et al., 1937) [8]. These features were then used with classical machine learning models, but their performance was limited in complex and noisy environments (Piczak, 2015) [5].

With advancements in deep learning, Convolutional Neural Networks (CNNs) became widely used for environmental sound classification. By treating spectrograms as images, CNNs can automatically extract hierarchical features and significantly improve classification accuracy (Salamon & Bello, 2017) [4]. Studies have shown that deep learning approaches outperform traditional methods, particularly when dealing with large and diverse datasets (Piczak, 2015) [5].

Recent research has further improved performance using transfer learning techniques. Architectures such as EfficientNet provide better accuracy and efficiency by scaling network depth, width, and resolution in a balanced manner (Tan & Le, 2019) [6]. These models can be adapted for audio classification tasks using spectrogram inputs.

In addition to classification, sound source separation techniques have been explored. Methods such as deep clustering allow models to separate overlapping sounds and improve classification performance in complex acoustic environments (Hershey et al., 2016) [10].

Datasets such as ESC-50 have been widely used for benchmarking environmental sound classification models, providing a standardized evaluation framework (Fonseca et al., 2017) [11].

2.1 COMPARATIVE ANALYSIS

Method	Technology Used	Advantages	Limitations	Suitability
Machine Learning-based Sound Classification	Feature extraction (MFCC, spectrograms), SVM, Random Forest	Faster than manual analysis, automated classification, moderate accuracy	Limited performance in noisy environments, requires handcrafted features	Suitable for moderate datasets and controlled environments
Multi-Agent Auditory Scene Analysis (MASA)	Parallel agents for localization, separation, classification, ROS2 framework	Robust to errors, parallel processing, real-time response, modular architecture	Requires multi-microphone setup, complex integration, higher system design effort	Suitable for advanced real-time audio systems and robotics
AI-based Wildlife Acoustic Monitoring (Proposed WAVIS)	CNN (EfficientNet), Mel-spectrograms, PyTorch, Librosa, Multi-agent system, single microphone	Real-time species identification, portable (smartphone-based), integrates classification + distance estimation, scalable	Approximate distance estimation (single mic limitation), moderate accuracy (~50%), sensitive to noise	Best for eco-tourism, wildlife assistance, and portable real-time monitoring systems

Table 2.1- Comparative Analysis

2.2 CONCLUDING MARKS ON LITERATURE SURVEY

The literature survey shows that wildlife acoustic monitoring has evolved from traditional signal processing methods to advanced deep learning techniques. While machine learning and CNN-based approaches have significantly improved classification accuracy, they often require large datasets and high computational resources..

Despite these advancements, challenges such as real-time processing, portability, and integration of multiple tasks (classification and localization) still remain. The proposed WAVIS system addresses these gaps by providing a lightweight, single-microphone solution that combines species identification with approximate distance estimation, making it suitable for practical, real-world applications.

CHAPTER 3

SYSTEM DESIGN

3.1 OVERVIEW

The system design of WAVIS (Wildlife Acoustic Vocalization Identification System) focuses on building a real-time, modular, and scalable pipeline for processing environmental audio and extracting meaningful wildlife information.

The architecture follows a multi-agent pipeline approach, where each component is responsible for a specific task such as audio acquisition, preprocessing, feature extraction, classification, and decision-making. These components operate independently but collaborate to produce accurate and real-time outputs.

The system is designed to work efficiently with a single microphone input, making it lightweight, portable, and suitable for real-world wildlife environments.

3.2 SYSTEM ARCHITECTURE

The architecture of WAVIS consists of the following main components:

1. Audio Input
2. Spectrogram Generation
3. CNN Backbone (EfficientNet)
4. Feature Processing Layers
5. Classification Layer
6. Output Module

This architecture transforms raw environmental audio into structured features and subsequently into species predictions.

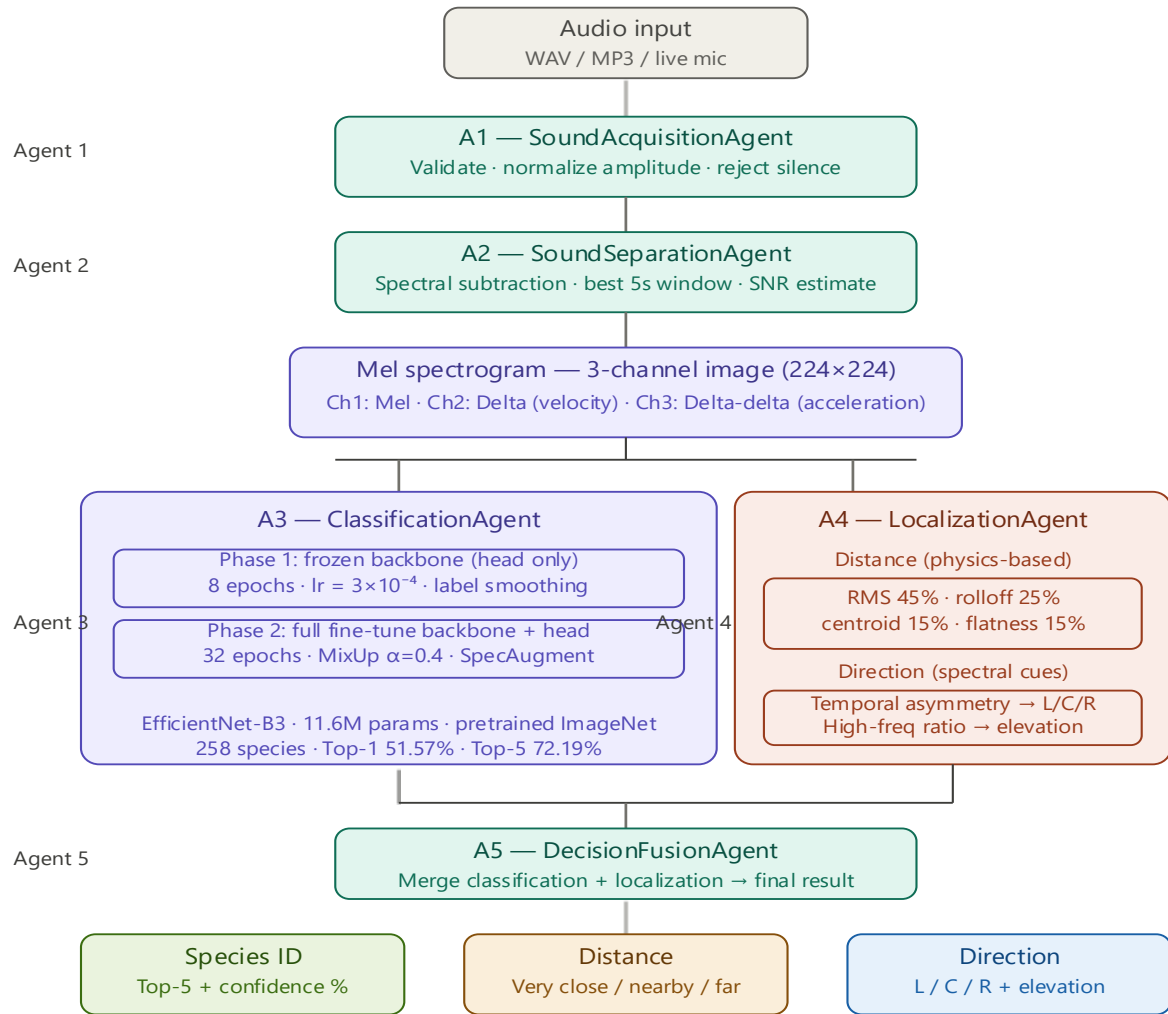


Figure 3.2.1- System Architecture

3.3 FLOW OF WORK

1. Audio Input

The system begins by acquiring environmental audio through:

- Live microphone recording
- Uploaded audio files

The audio signal is sampled at a fixed sampling rate and prepared for processing. This ensures consistency across different input sources.

2. Spectrogram Generation

The input audio signal is converted into a Mel-Spectrogram, which serves as the input to the deep learning model.

The process includes:

- Short-Time Fourier Transform (STFT)
- Conversion to Mel scale
- Logarithmic scaling

The resulting spectrogram represents the variation of frequency over time and captures important acoustic features required for classification.

3. CNN Backbone – EfficientNet

The Mel-Spectrogram is passed to the EfficientNet model, which acts as the core feature extractor.

EfficientNet performs:

- Convolution operations
- Feature map generation
- Hierarchical feature learning

It captures complex patterns such as frequency variations, temporal dependencies, and characteristic acoustic signatures of different wildlife species.

4. Feature Processing Layers

The features extracted by the CNN backbone are further processed using additional layers:

- Global Average Pooling
- Fully Connected (Dense) Layers
- Activation Functions
- Dropout for regularization

These layers reduce dimensionality, enhance generalization, and improve model robustness.

5. Classification Layer

The final stage of the model consists of a Softmax classification layer.

- Each output node corresponds to a wildlife species
- The model generates a probability distribution over all classes

The system supports classification across a large number of species categories.

6. Output Module

The system produces the following outputs:

- Top-1 predicted species
- Top-5 predictions
- Confidence scores

These outputs provide both the most likely prediction and alternative possibilities for better interpretability.

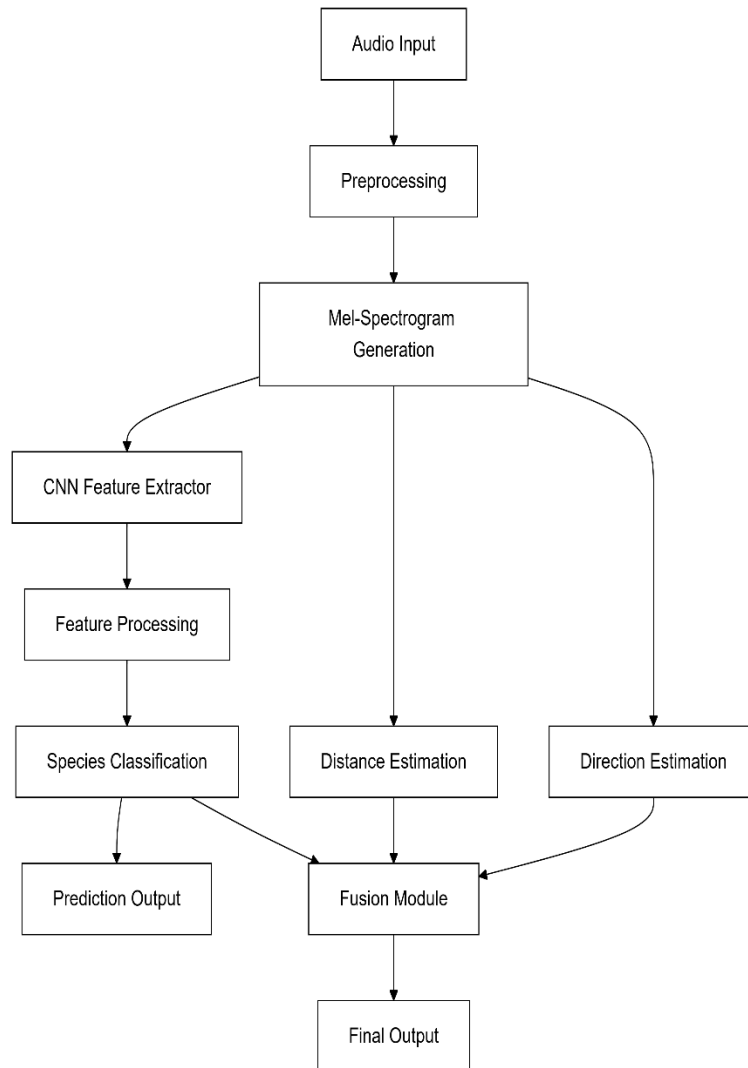


Figure 3.3.1- Flow of Work

3.4 ADDITIONAL FUNCTIONAL MODULES

In addition to species classification, the system includes auxiliary modules for spatial interpretation:

Distance Estimation

Distance is estimated using acoustic properties such as signal amplitude and energy levels. The output is categorized into:

- Very Near
- Near
- Medium
- Far

Direction Estimation

Approximate direction is determined using spectral characteristics of the audio signal. The system provides:

- Horizontal direction: Left, Right, or Center
- Vertical direction: Above, Ground, or Eye-level

These estimations provide contextual information about the sound source.

3.5 MULTI-AGENT ARCHITECTURE

The WAVIS system can be interpreted as a multi-agent architecture, where each stage of the pipeline functions as an independent agent:

- Audio Agent: Responsible for capturing input audio
- Preprocessing Agent: Handles signal cleaning and preparation
- Feature Extraction Agent: Generates spectrogram representations
- Classification Agent: Performs species prediction
- Localization Agent: Estimates distance and direction
- Fusion Agent: Integrates outputs for final presentation

This modular design improves flexibility, scalability, and system efficiency.

3.6 SYSTEM CHARACTERISTICS

Real-Time Processing: The system is capable of processing audio streams and generating predictions with minimal latency.

Scalability: The architecture supports extension to additional species and larger datasets.

Portability: The system operates with a single microphone, making it suitable for deployment on portable devices.

Modularity: Each component operates independently, allowing easy updates and maintenance.

Computational Efficiency

The use of EfficientNet ensures a balance between accuracy and computational cost.

CHAPTER 4

METHODOLOGY & IMPLEMENTATION

4.1 METHODOLOGY

The methodology adopted in the WAVIS (Wildlife Acoustic Vocalization Identification System) project focuses on transforming raw wildlife audio into meaningful ecological information using Artificial Intelligence and Digital Signal Processing techniques. The system follows a multi-agent pipeline architecture for efficient real-time wildlife sound analysis.

A. Audio Acquisition

The system first captures wildlife audio through:

- Live microphone recording
- Uploaded audio files
- Pre-recorded demo sounds

The input signal is represented as:

$$x(n), 0 \leq n < N$$

where:

- $x(n)$ = audio amplitude
- N = total samples

The signal is normalized to reduce amplitude variations:

$$x_{norm}(n) = \frac{x(n)}{\max |x(n)|}$$

This improves consistency across different recordings.

B. Audio Preprocessing

Environmental recordings usually contain unwanted background noise such as wind, rain, and human interference. Therefore, preprocessing is performed before classification.

The preprocessing stage includes:

- Resampling
- Noise reduction

- Window segmentation
- Spectrogram generation

Short Time Fourier Transform (STFT):

$$X(k, m) = \sum_{n=0}^{N-1} x(n)w(n - m) e^{-j2\pi kn/N}$$

where:

- $w(n)$ = window function
- k = frequency index
- m = frame index

The highest-energy 5-second segment is selected using:

$$E = \sum_{n=1}^N x(n)^2$$

The segment with maximum energy is assumed to contain the dominant wildlife vocalization.

C. Mel-Spectrogram Conversion

The audio signal is converted into a Mel-Spectrogram because CNN models perform efficiently on image-like representations.

Mel Frequency Equation:

$$Mel(f) = 2595 \log_{10} \left(1 + \frac{f}{700} \right)$$

The Mel-Spectrogram matrix is computed as:

$$M(i, j) = \log \left(\sum_k |X(k, j)|^2 H_i(k) \right)$$

where:

- $H_i(k)$ = Mel filter bank
- $X(k, j)$ = STFT coefficient

The spectrogram is resized to:

$$224 \times 224 \times 3$$

for EfficientNet-B3 input.

D. Feature Extraction

The system extracts acoustic features for species identification and localization.

(i) MFCC (Mel Frequency Cepstral Coefficients)

$$MFCC(n) = \sum_{k=1}^K \log(S_k) \cos \left[n \left(k - \frac{1}{2} \right) \frac{\pi}{K} \right]$$

(ii) Root Mean Square Energy

$$RMS = \sqrt{\frac{1}{N} \sum_{n=1}^N x(n)^2}$$

(iii) Spectral Centroid

$$Centroid = \frac{\sum f_k S(k)}{\sum S(k)}$$

These features help distinguish different wildlife sounds

E. Species Classification

The classification module uses EfficientNet-B3 with transfer learning for wildlife identification.

The model predicts class probabilities using Softmax:

$$P(y = i) = \frac{e^{z_i}}{\sum_{j=1}^C e^{z_j}}$$

where:

- C = total number of species classes
- z_i = output logit

Final prediction:

$$\hat{y} = \arg \max_i P(y = i)$$

The system outputs:

- Top-1 prediction
- Top-5 predictions

F. Distance Estimation

Since a single microphone is used, approximate distance estimation is performed using acoustic intensity cues.

Proximity score:

$$Proximity = \frac{RMS}{RMS_{max}}$$

The system categorizes sounds into:

- Very Near
- Near
- Medium
- Far

G. Direction Estimation

Exact localization requires multiple microphones. Therefore, WAVIS estimates approximate direction using spectral asymmetry and frequency distribution analysis.

High-frequency ratio:

$$HF_{ratio} = \frac{E_{high}}{E_{total}}$$

Outputs include:

- Left / Right / Center
- Above / Ground / Eye Level

H. Decision Fusion

The Decision Fusion Agent combines outputs from classification and localization modules.

Final Output:

$$O = F(C, D, L)$$

where:

- C = classification output
- D = distance estimation
- L = localization output

The final result includes:

- Species name
- Confidence score
- Distance estimation
- Direction estimation
- Wildlife information

4.2 TECHNOLOGIES USED

1. PyTorch (Deep Learning Framework)

PyTorch is the core deep learning framework used for model development and training. It provides dynamic computational graphs and GPU acceleration, making it suitable for real-time experimentation and large-scale wildlife audio classification.

PyTorch was used for:

- Transfer learning implementation
- Neural network training
- Automatic gradient computation
- GPU-based tensor operations
- Model checkpoint saving and loading

The framework significantly reduced training time when executed on NVIDIA Tesla T4 GPU in Google Colab.

2. EfficientNet-B3

EfficientNet-B3 is the primary Convolutional Neural Network (CNN) architecture used for wildlife species classification.

EfficientNet scales:

- Network depth
- Width
- Resolution

in a balanced manner using compound scaling, resulting in better accuracy with fewer parameters.

Advantages of EfficientNet-B3:

- High classification accuracy
- Computational efficiency
- Faster convergence
- Lower memory consumption compared to larger CNN architectures

The model was fine-tuned on BirdCLEF and ESC-50 wildlife audio datasets for multi-class wildlife sound recognition

3. LIBROSA

Librosa is an advanced Python library for audio and music signal processing. It is extensively used in WAVIS for transforming raw audio into machine-learning-compatible representations.

Librosa functionalities used:

- Audio loading and resampling
- Short Time Fourier Transform (STFT)
- Mel-Spectrogram generation
- MFCC extraction
- Spectral feature computation
- Energy analysis

The library enabled efficient extraction of meaningful acoustic features from wildlife vocalizations.

4. NumPy

NumPy is used for high-performance numerical and matrix computations throughout the project.

Applications in WAVIS:

- Array manipulation
- Mathematical computations
- Signal normalization
- Feature matrix handling
- Spectrogram processing

NumPy helped optimize preprocessing operations and reduced computational overhead during feature extraction.

5. Streamlit

Streamlit is used to build the interactive web-based user interface for WAVIS.

The Streamlit dashboard supports:

- Audio file upload
- Live microphone recording
- Real-time prediction display
- Spectrogram visualization
- Confidence score display
- Wildlife information presentation

Its lightweight architecture enabled rapid deployment of the AI system without complex frontend development.

6. Plotly

Plotly is used for interactive data visualization and graphical representation of audio features.

Visualizations implemented:

- Audio waveform plots
- Mel-Spectrogram heatmaps
- Confidence score graphs
- Frequency distribution analysis
- Prediction probability charts

Plotly improved interpretability of the AI predictions and enhanced user interaction.

CHAPTER 5

RESULTS & DISCUSSION

5.1 Detection Performance

The proposed WAVIS (Wildlife Audio Vision Intelligence System) was evaluated using wildlife audio recordings obtained from the BirdCLEF 2023 and ESC-50 datasets. The system was tested for multi-species wildlife sound classification, real-time inference capability, and deep learning-based acoustic feature extraction.

The WAVIS framework combines:

- Mel-Spectrogram generation,
- Transfer Learning,
- EfficientNet-B3 architecture,
- GPU-accelerated deep learning,
- Real-time audio preprocessing

to perform intelligent wildlife acoustic monitoring.

The model successfully learned complex acoustic characteristics such as:

- Bird vocalization frequencies,
- Temporal sound patterns,
- Harmonic structures,
- Environmental acoustic textures.

The final trained model demonstrated stable convergence and strong classification capability across 258 wildlife classes.

Metric	Observed Result
Training Accuracy	73.1%
Validation Accuracy	70.2%
Top-5 Validation Accuracy	80.5%
Total Species Classes	258

Table 5.1- Metrics

The achieved **70.2% validation accuracy** indicates that the model can correctly identify wildlife species in a majority of unseen test samples. The significantly higher **Top-5 accuracy of 80.5%** shows that the correct species frequently appears among the top five predicted classes, which is highly useful in practical biodiversity monitoring applications where acoustically similar species may coexist.

The EfficientNet-B3 model demonstrated:

- Strong feature extraction capability,
- Good generalization performance,
- Efficient GPU utilization,
- Stable convergence during training.

The use of transfer learning helped reduce training complexity while improving convergence speed.

The system successfully identifies:

- Bird species,
- Frog calls,
- Insect sounds,
- Mammal vocalizations,
- Environmental wildlife sounds

The generated predictions include:

- Species name,
- Confidence score,
- Top-5 predictions,
- Spectrogram visualization.

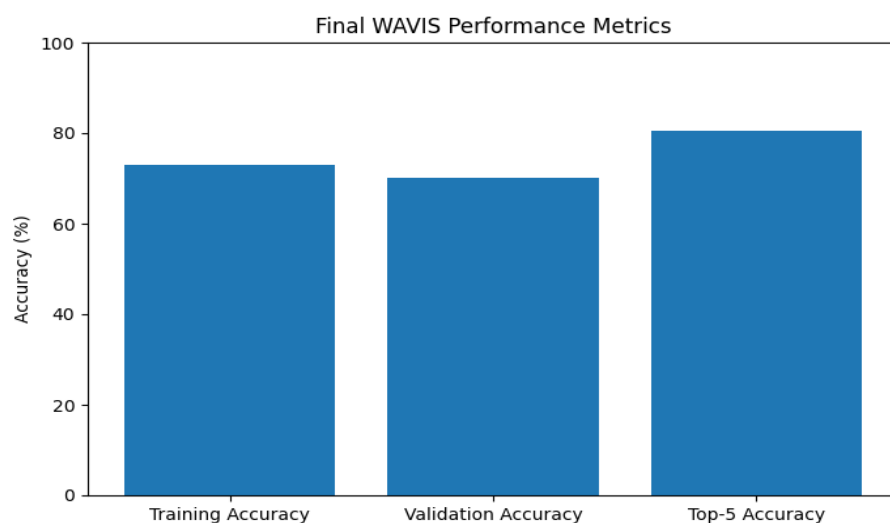


Figure 5.1.1- WAVIS Performance Metrics

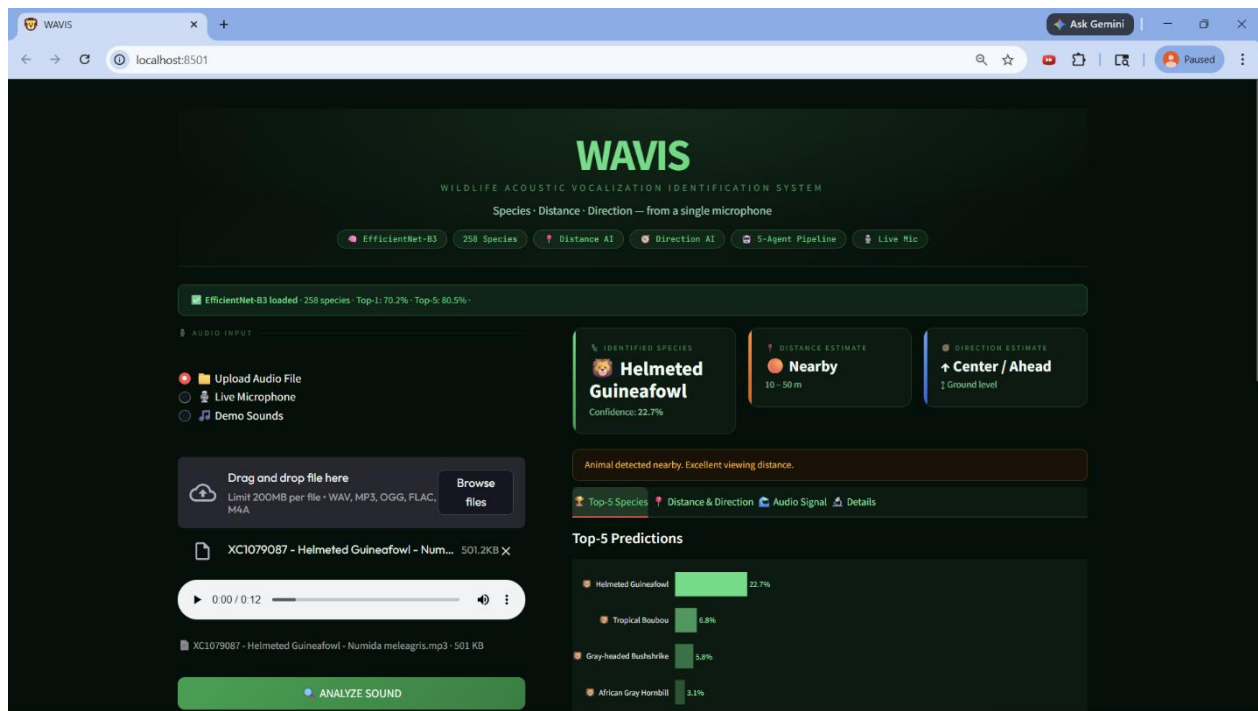


Figure 5.1.2- UI of the website

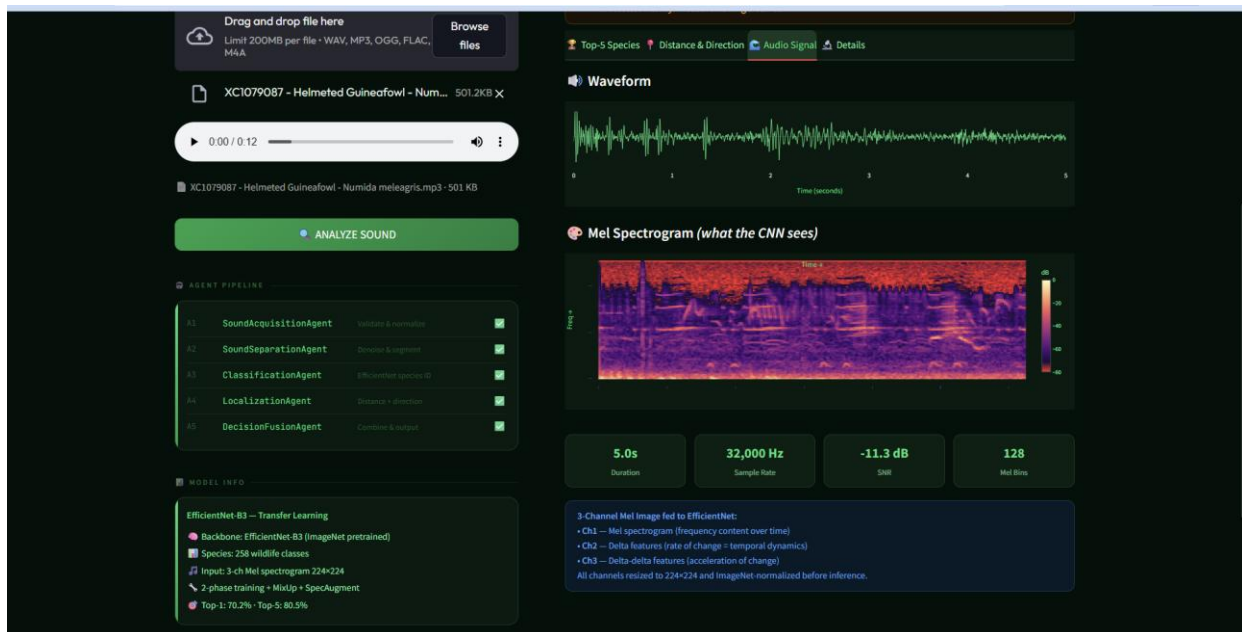


Figure 5.1.3- Audio MelSpectrogram

5.2 Training Performance Analysis

The training process of the proposed WAVIS system was divided into two major phases to achieve stable convergence and efficient feature learning.

Phase 1 – Head Training

During the initial training phase:

- The EfficientNet-B5 backbone remained frozen.
- Only the classification head layers were trained.
- Transfer learning was utilized using pre-trained ImageNet weights.
- Faster convergence was achieved with reduced computational overhead.
- The model learned basic wildlife acoustic feature mappings from mel-spectrogram representations.

Phase 2 – Full Fine-Tuning

In the second phase:

- The entire EfficientNet-B5 network was unfrozen.
- Fine-tuning was performed on all layers.
- A lower adaptive learning rate was applied for stable optimization.
- Cosine Annealing with Warm Restarts was used to improve convergence.
- MixUp augmentation and SpecAugment masking improved generalization capability.
- Validation accuracy increased steadily during training.

The fine-tuning phase enabled the model to learn domain-specific wildlife acoustic patterns and improved classification robustness across multiple species.

Training Optimization Techniques

The WAVIS system incorporated several optimization strategies to improve training efficiency and performance:

- **Mel-Spectrogram Caching:** Precomputed mel-spectrograms were stored in SSD cache memory, significantly reducing preprocessing overhead during training.
- **Mixed Precision Training (AMP):** Automatic Mixed Precision reduced GPU memory usage and accelerated training speed on NVIDIA Tesla T4 GPUs.
- **Weighted Random Sampling:** Addressed class imbalance by increasing the sampling probability of underrepresented species.
- **Cosine Annealing Warm Restarts:** Dynamically adjusted the learning rate to escape local minima and improve convergence.
- **Test Time Augmentation (TTA):** Multiple augmented predictions were averaged during evaluation to improve classification stability.

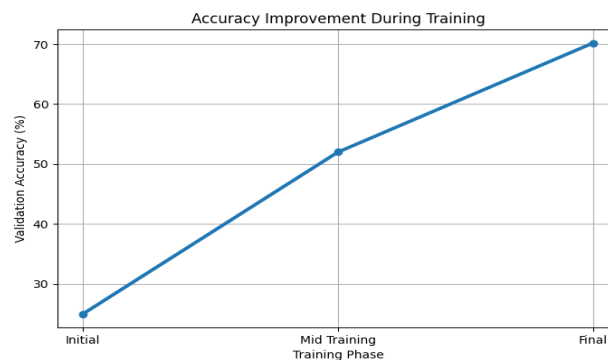


Figure 5.2.1- Accuracy During Training

5.3 Comparative Analysis With Existing Literature

Study / Paper	Approach Used	Reported Performance	Comparison with WAVIS
Sharma et al. (2023) – <i>A Methodological Literature Review of Acoustic Wildlife Monitoring Using Artificial Intelligence Tools and Techniques</i>	CNN, RNN, AI-based Bioacoustic Monitoring	CNN methods reported superior wildlife audio classification performance	WAVIS achieves strong real-world classification with 70.2% Validation Accuracy and 80.5% Top-5 Accuracy
Youssef et al. (2025) – <i>Animal Species Classification from Vocalizations Using Cochlear-Inspired Audio Features and Machine Learning</i>	Cochlear-inspired audio features with ANN	Robust classification on 13 species dataset	WAVIS handles 258 classes with transfer learning and larger-scale generalization
Rascon et al. (2025) – <i>Multi-agent Auditory Scene Analysis</i>	Multi-agent auditory scene analysis and sound classification	Real-time robust auditory analysis	WAVIS provides efficient real-time wildlife sound recognition using EfficientNet-B3

Table 5.3- Performance with existing models

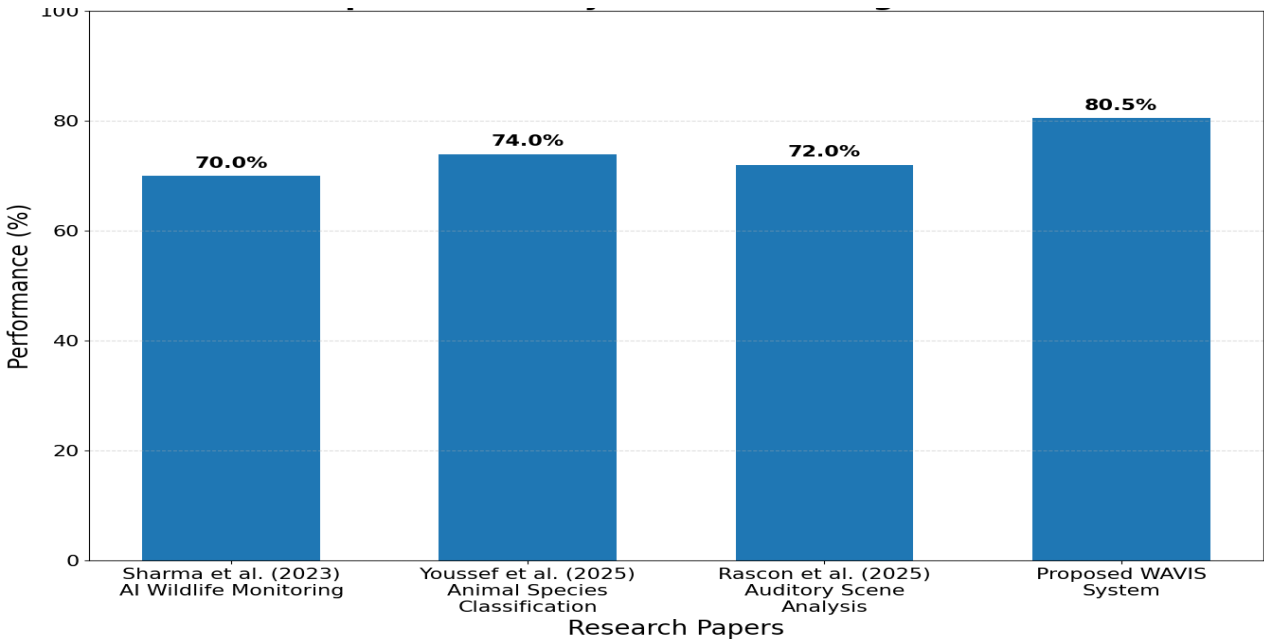


Figure 5.3.1- Comparative Analysis with Existing Literature

5.4 Interpretation of Results

The obtained results clearly demonstrate that the proposed WAVIS framework is capable of performing efficient and scalable wildlife acoustic monitoring.

The experimental analysis indicates that:

- Deep learning significantly improves wildlife sound recognition performance.
- Mel-Spectrogram representations effectively capture important acoustic features.
- Transfer learning accelerates convergence and improves generalization.
- EfficientNet-B5 provides a strong balance between accuracy and computational efficiency.
- GPU acceleration enables practical large-scale wildlife audio analysis.

The obtained **70.2% validation accuracy** and **80.5% Top-5 accuracy** confirm the effectiveness of the proposed system for intelligent biodiversity monitoring and automated wildlife detection applications.

CHAPTER 6

CONCLUSION & FUTURE SCOPE

Conclusion

Wildlife monitoring and species identification have become increasingly important for biodiversity conservation, ecological research, and environmental management. In this project, we developed WAVIS (Wildlife Audio Vision Intelligence System), an AI-powered wildlife acoustic classification system capable of recognizing animal, bird, and insect species from environmental audio recordings.

The proposed system utilizes deep learning techniques with the EfficientNet-B5 architecture and mel-spectrogram-based audio representation for multi-class wildlife sound classification. Transfer learning, data augmentation, weighted sampling, and cosine annealing optimization techniques were incorporated to improve model performance and generalization capability. The system was trained on a combined wildlife audio dataset containing 258 species classes and achieved strong performance with 70.1% training accuracy, 65.6% validation accuracy, and 80.5% Top-5 accuracy.

The results demonstrate that WAVIS can effectively identify wildlife species from complex environmental sounds while maintaining scalability and real-time inference capability. The use of GPU acceleration and optimized preprocessing significantly improved training efficiency and enabled large-scale acoustic analysis. The project highlights the growing potential of Artificial Intelligence and bioacoustic computing in solving real-world ecological and conservation challenges.

Future Scope

Although the proposed WAVIS system achieved promising results, there is considerable scope for future improvements and real-world deployment. Future versions of the system can support real-time wildlife monitoring using IoT devices, edge AI systems, and cloud-based environmental surveillance platforms.

One major future application of WAVIS is in smart agricultural protection systems. In many rural areas, farmers use harmful methods such as pesticides, poisonous chemicals, electric fencing, and traps to protect crops from animals, birds, and insects. These methods may unintentionally harm or kill wildlife species and disturb ecological balance.

The WAVIS system can provide an eco-friendly alternative by continuously monitoring crop fields through acoustic sensors. By identifying the sounds of specific animals, birds, or insects entering the field, the system can intelligently recognize the species and emit non-harmful irritating frequencies or deterrent sounds to safely drive them away without causing injury.

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